

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

**Problem 1.** (5 points) Find a particular solution for the following differential equation using the method of undetermined coefficients:

$$y^{(4)} + 4y'' = t^2$$

Solution: 1. Solve the homogeneous equation for  $y_c(t)$ .

$$\gamma^4 + 4\gamma^2 = 0$$

$$\gamma^2(\gamma^2 + 4) = 0 \Rightarrow \gamma_1 = \gamma_2 = 0, \gamma_3 = 2i, \gamma_4 = -2i$$

$$\Rightarrow y_c(t) = C_1 + C_2 t + C_3 \cos 2t + C_4 \sin 2t$$

2. We guess a suitable form for  $Y(t)$  as

$$Y(t) = t^s(A + Bt + Ct^2)$$

$$\stackrel{s=2}{=} At^2 + Bt^3 + Ct^4$$

$$Y'(t) = 2At + 3Bt^2 + 4Ct^3$$

$$Y''(t) = 2A + 6Bt + 12Ct^2$$

$$Y'''(t) = 6B + 24Ct$$

$$Y^{(4)}(t) = 24C$$

$$\begin{aligned} \Rightarrow Y^{(4)} + 4Y'' &= 24C + 4(2A + 6Bt + 12Ct^2) \\ &= 48Ct^2 + 24Bt + 8A + 24C \\ &= t^2 \end{aligned}$$

$$\Rightarrow \begin{cases} 48C = 1 \\ 24B = 0 \\ 8A + 24C = 0 \end{cases} \Rightarrow \begin{aligned} C &= 1/48 \\ B &= 0 \\ A &= -1/16 \end{aligned}$$

$$\Rightarrow Y(t) = -\frac{t^2}{16} + \frac{t^4}{48}$$

**Problem 2.** (5 points) Find a particular solution for the following differential equation using the method of variation of parameters:

$$y''' - y' = t$$

*Solution:* Characteristic equation

$$r^3 - r = 0 \Rightarrow r(r^2 - 1) = 0 \Rightarrow r(r-1)(r+1) = 0$$

$$\Rightarrow r_1 = 0, r_2 = 1, r_3 = -1$$

$$\Rightarrow y_1(t) = 1, y_2(t) = e^t, y_3(t) = e^{-t}$$

$$W(y_1, y_2, y_3)(t) = \begin{vmatrix} 1 & e^t & e^{-t} \\ 0 & e^t & -e^{-t} \\ 0 & e^t & e^{-t} \end{vmatrix} = 1 \cdot \begin{vmatrix} e^t & -e^{-t} \\ e^t & e^{-t} \end{vmatrix} = 2$$

$$W_1(t) = \begin{vmatrix} 0 & e^t & e^{-t} \\ 0 & e^t & -e^{-t} \\ 1 & e^t & e^{-t} \end{vmatrix} = \begin{vmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{vmatrix} = -2$$

$$W_2(t) = \begin{vmatrix} 1 & 0 & e^{-t} \\ 0 & 0 & -e^{-t} \\ 0 & 1 & e^{-t} \end{vmatrix} = \begin{vmatrix} 0 & -e^{-t} \\ 1 & e^{-t} \end{vmatrix} = e^{-t}$$

$$W_3(t) = \begin{vmatrix} 1 & e^t & 0 \\ 0 & e^t & 0 \\ 0 & e^t & 1 \end{vmatrix} = \begin{vmatrix} e^t & 0 \\ e^t & 1 \end{vmatrix} = e^t$$

$$\begin{aligned} \Rightarrow Y(t) &= y_1(t) \int \frac{g(t) W_1(t)}{W(t)} dt + y_2(t) \int \frac{g(t) W_2(t)}{W(t)} dt + y_3(t) \int \frac{g(t) W_3(t)}{W(t)} dt \\ &= 1 \cdot \int \frac{t(-2)}{2} dt + e^t \int \frac{t e^{-t}}{2} dt + e^{-t} \int \frac{t e^t}{2} dt \\ &= -\frac{t^2}{2} + \frac{e^t}{2} [-(t+1)e^{-t}] + \frac{e^{-t}}{2} (t-1)e^t \\ &= -\frac{t^2}{2} - 1 \end{aligned}$$